

AUGUST 10, 2026

ACCELERATOR & HBM

AI NETWORKING MODEL

CORE RESEARCH

Rubin Ultra NVL576 Architecture: Flash Overview

3 MINUTES

By Tony Chien, Wega Chu and Myron Xie

Although the exact spec and architecture design of Rubin Ultra remains in flux, the design and spec should be finalized soon. At the current stage, we have discussed the HBM despecing and the offering of multiple SKUs running at different TDPs at the GPU level. At the system level, we have also highlighted

the use of NPO to achieve a bigger scale up world size of 576 Rubin Ultra packages. In this note, we will compare the spec between Rubin and that of what we currently view Rubin Ultra as (subject to change later). Then, we will focus on the new Oberon architecture being proposed to enable the NVL72 cross rack scale up.

Nvidia Roadmap								
	2024	2025		2026		2027		
Chip and Package Level								
Generation	Blackwell			Rubin				
Accelerator	B200/ GB200	B300/GB300 (Ultra)	B300 (single die, B300A)	Rubin	LP30/LP35	Rubin Ultra		
GPU TDP (W)	700/1,200	1,100/1,400	600	2,300	~600	1,800/2,600		
Foundry Node	4NP			N3P (3NP)	SF4X	N3P (3NP)		
Logic Die Configuration	2 x Reticle Sized GPU			2 x Reticle Sized GPU, 1 C2C I/O Die, 1 NVLink I/O Die	1 x Reticle Sized LPU	2 x Reticle Sized GPU, 1 NVLink I/O Die		
FP4 PFLOPs - Dense (per Package)	10	15	4.6	35 ⁽²⁾	-	35		
FP8 PFLOPs - Dense (per Package)	5	5	4.6	17.5	1.2	17.5		
FP16 PFLOPs - Dense (per Package)	2.5	2.5	4.6	4	0.6	4		
Memory (per Package)	192GB HBM3E	288GB HBM3E	144GB HBM3E	288GB HBM4	500MB SRAM	192GB HBM4		
HBM Stacks	8		4	8	-	8		
Memory Bandwidth	8TB/s		4TB/s	20TB/s	150TB/s	21TB/s		
Packaging	CoWoS-L			CoWoS-L	FC-BGA	CoWoS-L		
SerDes speed (Gb/s uni-di)	224G			224G Bi-d ⁽³⁾	112G	224G Bi-d ⁽³⁾		
System Level and Form Factor								
System	NVL72		NVL16	Rubin NVL8 HGX	Vera Rubin NVL72	Nvidia Groq 3 LPX	Vera Rubin NVL72	Vera Rubin Ultra NVL576
Form Factor	Oberon		HGX	HGX	Oberon	4 racks x 256 Groq 3 LPUs (4)	Oberon	8x Oberon Racks
# of Logical GPUs	72	72	16	8	72	1,024	72	576
# of GPU dies / compute chiplets	144	144	16	16	144	1,024	144	1,152
CPU	Grace		Intel or AMD	N/A	Vera	Intel Granite Rapids	Vera	
# of CPU Sockets	36	36	16	2	36	128 FPGAs 64 CPUs	36	288
Scale up links	Copper Backplane		UBB (PCB)	UBB (PCB)	Copper Backplane	PCB (Intra-node) + Copper Backplane (Inter-node) + AEC/Optics (Inter-rack)	Copper Backplane	Within Rack: Copper Backplane Between Racks: Dragonfly NPO/CPCs

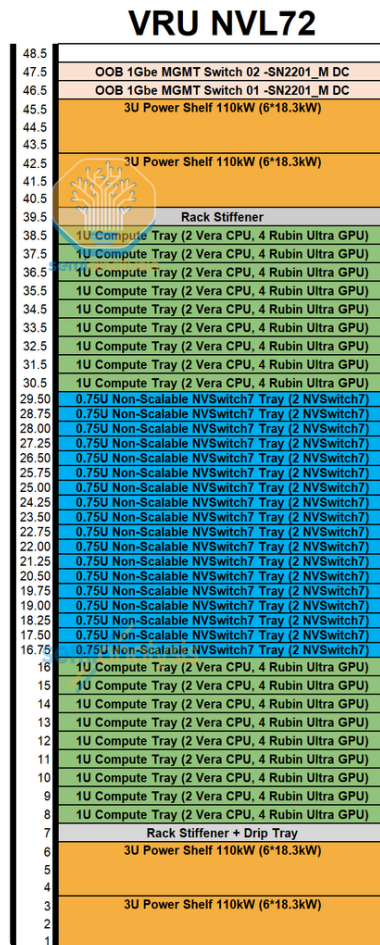
Source: SemiAnalysis, Nvidia

Nvidia Roadmap								
	2024		2025		2026			
Scale-Up Networking								
Generation	Blackwell				Rubin			
Accelerator	B200/GB200	B300/GB300 (Ultra)	B300 (single die, B300A)	Rubin		LP30/LP35	Rubin Ultra	
System	NVL72		NVL16	Rubin NVL8 HGX	Vera Rubin NVL72	1,024 Nvidia Groq 3 LPUs	Vera Rubin Ultra NVL72	Vera Rubin Ultra NVL576
Form Factor	Oberon		HGX	HGX	Oberon	4 racks x 256 Groq 3 LPUs (4)	Oberon	8x Oberon Racks
# of Logical GPUs	72	72	16	8	72	1,024	72	576
# of GPU dies	144	144	16	16	144	1,024	144	1,152
NVLink Generation	NVLink 5			NVLink 6		Groq C2C	NVLink 7	
Scale-Up Topology	One-layer Switched All-to-All			One-Layer Switched All-to-All	One-Layer Switched All-to-All	Dragonfly	One-Layer Switched All-to-All	Direct Connect NPO
Within Rack Scale-Up	Copper Backplane		UBB (PCB)	UBB (PCB)	Copper Backplane	PCB within Tray Copper Backplane between Trays	Copper Backplane	Copper Backplane
Between Rack Scale-Up	N/A			N/A		AEC/Pluggable Optics	N/A	NPO/CPO
Bandwidth per Logical GPU (GB/s uni-di)	900			1,800		1,250	1,800	
Lane Speed (Gb/s uni-di)	200G PAM4			400G (200G PAM4 Simultaneous Bidi)		100G	400G (200G PAM4 Simultaneous Bidi)	
NVLink Switch Generation	NVLink 5 Switch			NVLink 6 Switch		N/A	NVLink 7 Switch	
Switch Aggregate BW (GB/s uni-di)	3,600			3,600		N/A	3,600	

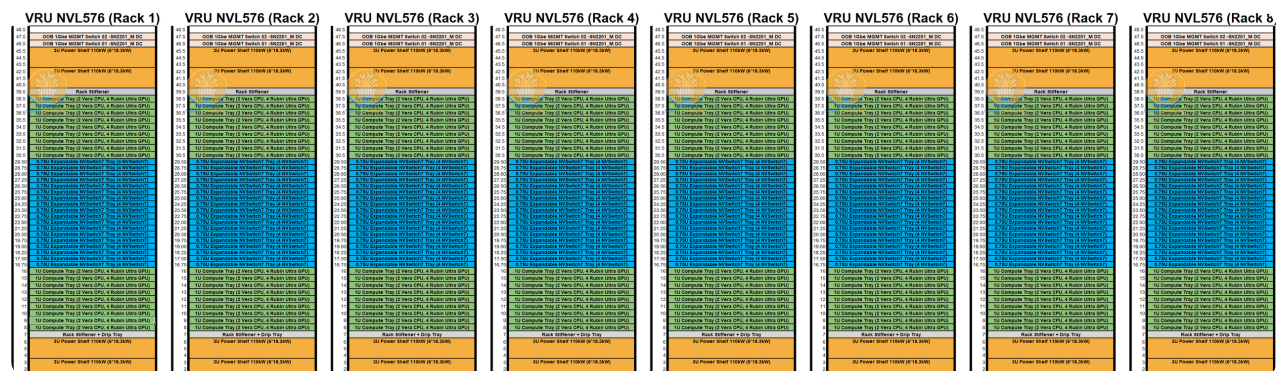
Source: SemiAnalysis, Nvidia

At the system level, we can split the architecture into three sections: compute tray, NVLink Switch tray, and the rack elevation diagram. The compute tray design of Rubin Ultra (Oberon) will be mostly the same as that of Rubin. The

only upgrade we are observing is on the Tachyon HPM board PCB layers from 26 to 30 layers. This upgrade is to accommodate a larger TDP from Rubin to Rubin Ultra (to support up to 2.6KW per Rubin Ultra GPU though we think mainstream will be 1.8KW). Also, the LPDDR5X SOCAMM will be placed on the backside of the HPM instead of the topside of the HPM board.



Source: SemiAnalysis,
Nvidia

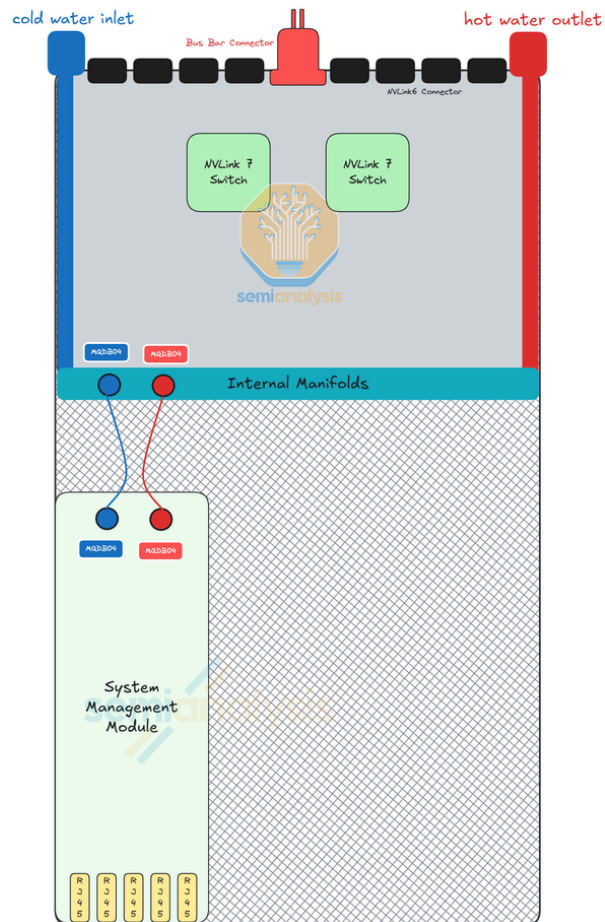


Source: SemiAnalysis, Nvidia

For the rack elevation, the changes are more obvious and significant. Currently, the Oberon rack (10+9+8) is arranged with 10 compute trays at the top, 9 NVLink Switch trays in the middle, and 8 compute trays at the bottom. Each of these trays is 1U in height. For the Oberon of Rubin Ultra (9+18+9), the amount of compute trays remains the same at 18 units, but it's split evenly between the top and the bottom. The NVLink Switch trays are doubled to 18 units per rack, and the height is reduced to 0.75U per tray. This is to shorten the distance between the furthest compute tray from the furthest NVLink Switch tray to ensure that the NVLink signal running on copper backplane can still drive through the channel properly. Before the longest distance between the compute tray and the NVLink Switch tray is 19U, now the longest distance would only increase slightly, despite doubling NVSwitch trays, to 22.5U.

Depending on the scale up world-size of NVL72 or NVL576, the NVLink Switch tray (codename: Portia) will be configured accordingly into the non-expandable version for NVL72, and the expandable version for NVL576. Within the expandable version, there are currently both NPO and CPO versions developing currently and we believe the NPO will be the go to market version given the relative maturity of the form factor compared to CPO. Note that the rack elevation for both non-expandable and expandable versions are the same, meaning that the only difference is in the NVLink Switch tray. The same rack elevation infrastructure design allows the backplane to be the same for both the expandable and the non-expandable version, easing the supply chain complexity on the backplane side.

VRU NVL72 NVSwitch 7 Tray Top View

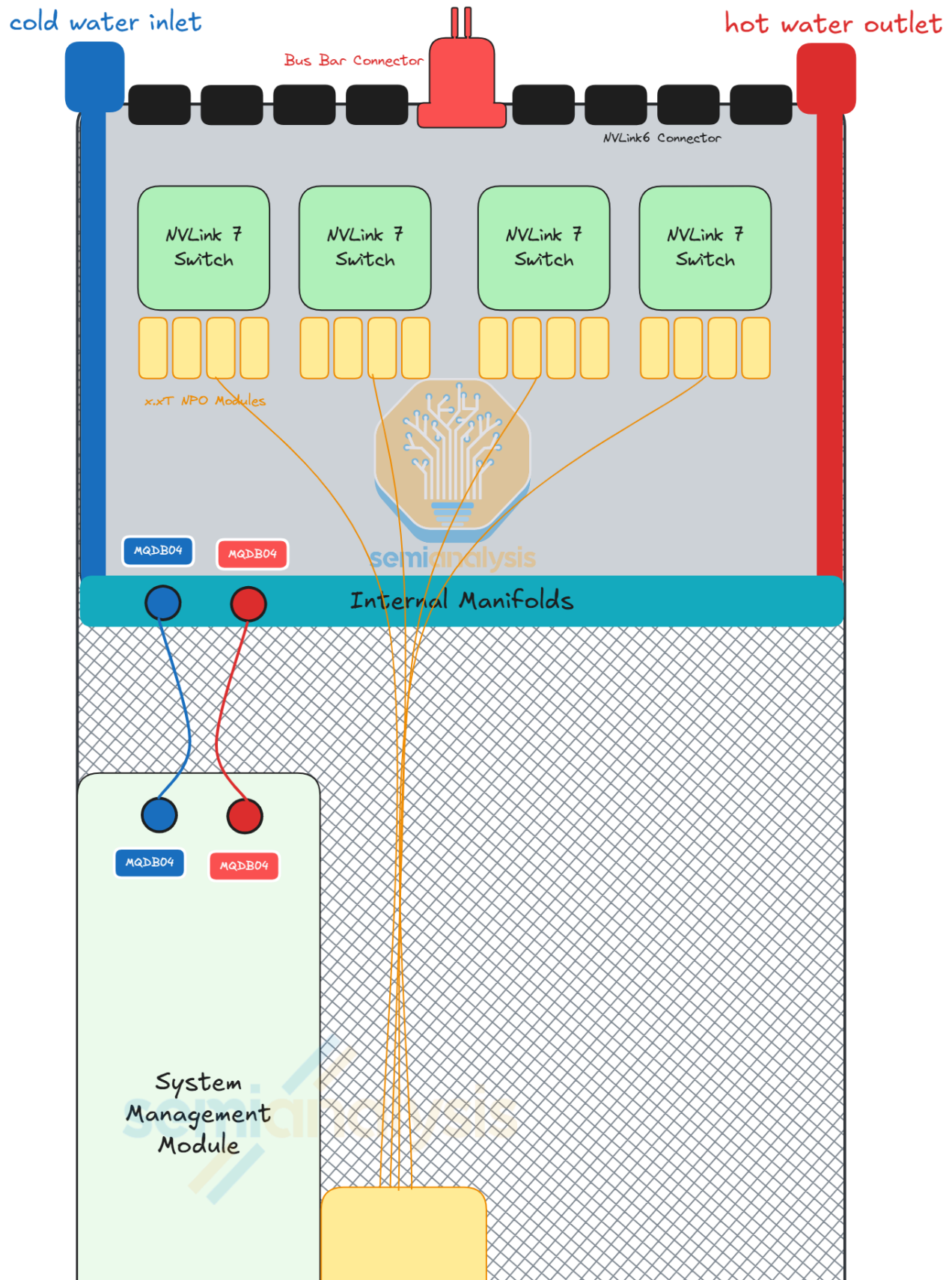


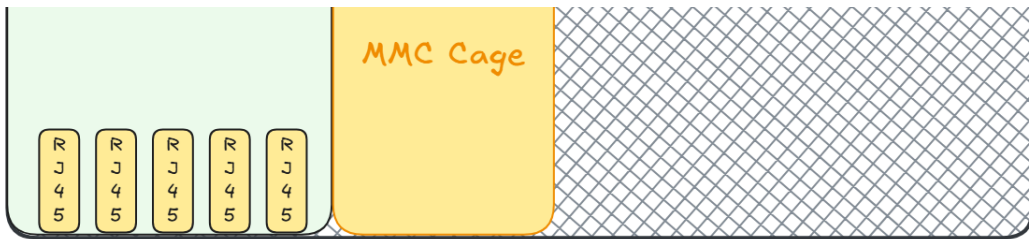
Source: SemiAnalysis, Nvidia

Non-expandable NVLink Switch tray will have 2 NVLink Switch ASICs per switch tray. With 18 NVLink Switch trays, this adds up to 36 NVLink Switch ASICs in total per rack, equalling that of Rubin Oberon. In other words, the scale up bandwidth between the non-expandable Vera Rubin Ultra Oberon and Vera Rubin Oberon is the same, and the amount of scale up NVLink Switch ASICs per rack is the same despite doubling the NVLink Switch trays.

VRU NVL576

NVSwitch 7 (NPO) Tray Top View

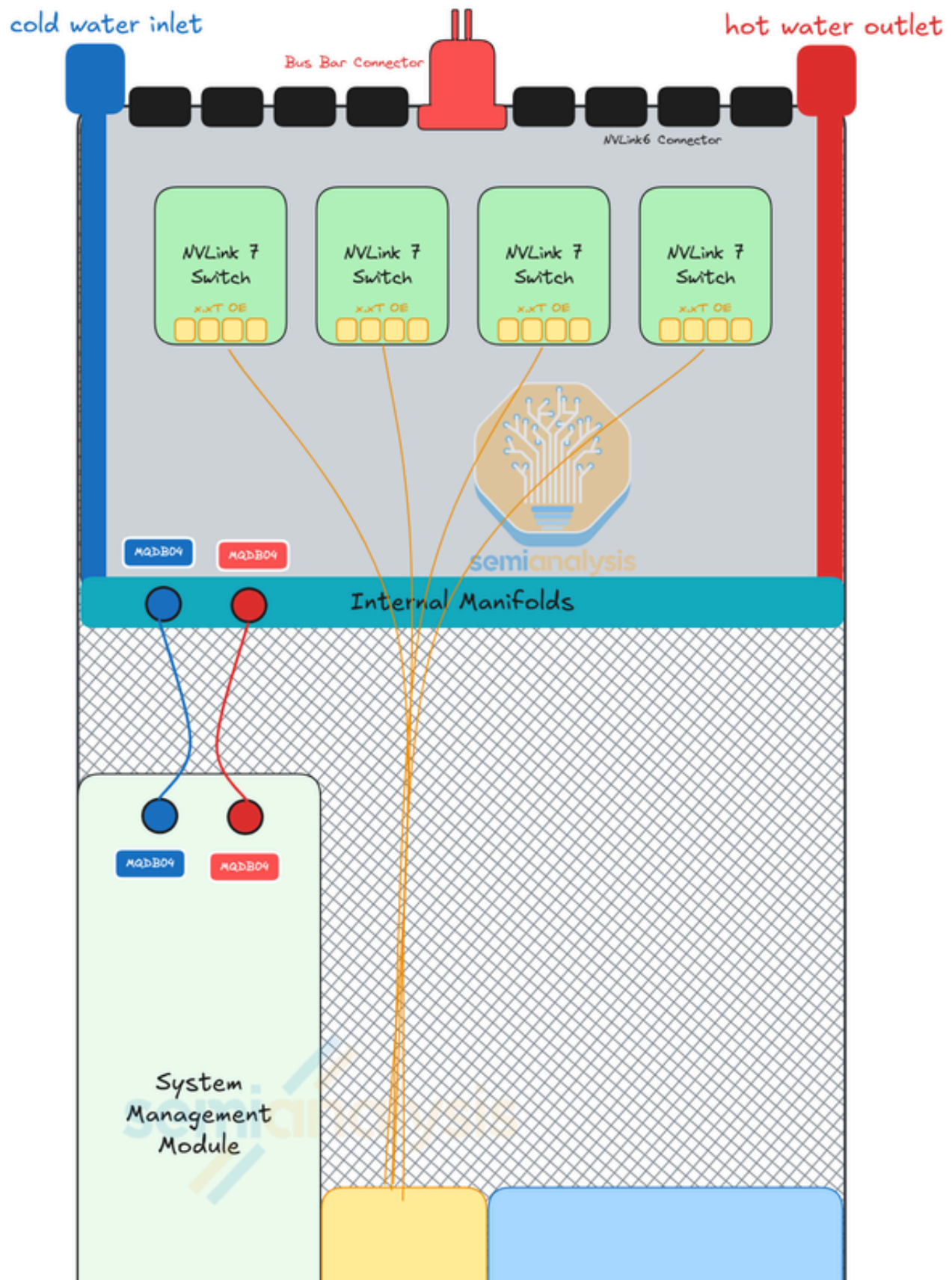




Source: SemiAnalysis, Nvidia

VRU NVL576

NVSwitch 7 (CPO) Tray Top View





Source: SemiAnalysis, Nvidia

Expandable NVLink Switch trays come in two variations, NPO and CPO. For both expandable versions, the number of NVLink Switch ASICs doubles to 4 compared to the non-expandable version. This means, there will be a total of 72 NVLink Switch ASICs per rack. The reason for this is to increase the total bandwidth to allow for cross rack connections. For the NPO version, the NPO modules will be socketed onto the PCB next to the NVLink Switch ASICs. For the CPO version, there would be 4 optical engines per NVLink Switch ASICs; however, they are not replaceable.

There are some nuanced implications to hardware and components for the new rack elevation and the non-expandable and expandable switches. The new rack elevation is positive for chassis and railkits suppliers for the NVSwitch as there are more NVLink Switch trays per rack. The backplane content will be up as PHD2 connector upgrades to PHD3, while the DP counts per rack remain the same. The PCB for the Tachyon HPM will be increased by 4 layers, while the material remains the same. The NPO participants will benefit from the NVL576 formfactor, which we have covered in the networking model note. Lastly, PCB for the NVLink Switch 7 tray – Portia will be different depending on the versions of the switch. The PCB content for the NPO version of Portia will be the most complex, while the PCB for the non-expandable will be complex yet halved in area compared to the expandable version.

Disclaimers

Analyst Certifications and Independence of Research.

Each of the analysts whose names appear in this report hereby certify that all the views expressed in this Report accurately reflect our personal views about any and all of the subject securities or issuers and that no part of our compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views in this Report. SemiAnalysis LLC (the "Company") is an independent equity research provider. The Company is not a member of the Financial Industry Regulatory Authority or the Securities Investor Protection Corporation, and it is not a registered broker dealer or investment adviser.

Private Investment Disclosure.

Each analyst and/or author contributing to this Report hereby discloses that they may hold, directly or indirectly, private investments in companies, sectors, or asset classes discussed or referenced herein. Such holdings, if any, are disclosed externally with a third-party vendor. The Company requires that no author or analyst allow their personal investment positions to influence the views, opinions, analyses, or conclusions expressed in this Report. Readers should be aware that the existence of such holdings, even where disclosed, may present a potential conflict of interest, and are encouraged to weigh this accordingly when evaluating the contents of this Report.

Limitation of Research and Information.

This Report has been prepared for distribution to only qualified institutional or professional clients of the Company. The contents of this Report represent the views, opinions, and analyses of its authors. The information contained herein does not constitute financial, legal, tax or any other advice. All third-party data presented herein were obtained from publicly available sources which are believed to be reliable; however, the Company makes no warranty, express or implied, concerning the accuracy or completeness of such information. In no event shall the Company be responsible or liable for the correctness of, or update to, any such material, or for any direct, indirect, incidental or consequential loss, damage or lost opportunities (including loss of profits, revenue or goodwill) resulting from use of this data or from any investment decisions based on information or research obtained from the Company.

Nothing contained in this Report or any distribution by the Company should be construed as any offer to sell, or any solicitation of an offer to buy, any security or investment. Any research or other material received should not be construed as individualized investment advice. Investment decisions should be made as part of an overall portfolio strategy and you should consult with a professional financial advisor, legal and tax advisor prior to making any investment decision.

Reproduction and Distribution Strictly Prohibited.

No user of this Report may reproduce, modify, copy, distribute, sell, resell, transmit, transfer, license, assign or publish the Report itself or any information contained therein. Notwithstanding the foregoing, clients with access to working models are permitted to alter or modify the information contained therein, provided that it is solely for such client's own use. This Report is not intended to be available or distributed for any purpose that would be deemed unlawful or otherwise prohibited by any local, state, national or international laws or regulations or would otherwise subject the Company to registration or regulation of any kind within such jurisdiction.

Copyrights, Trademarks, Intellectual Property.

The Company, and any logos or marks included in this Report are proprietary materials. The use of such terms and logos and marks without the express written consent of the Company is strictly prohibited. The copyright in the pages or in the screens of the Report, and in the information and material therein, is proprietary material owned by the Company unless otherwise indicated. The unauthorized use of any material on this Report may violate numerous statutes, regulations and laws, including, but not limited to, copyright, trademark, trade secret or patent laws.

ETF and Regulatory Status Disclaimer.

From time to time, The Company may collaborate with asset managers, issuers, sponsors, index providers or other financial institutions in connection with the development, launch or support of exchange-traded funds ("ETFs") and other investment products, including by providing research, market analysis, index-related research or other research related services. The Company does not

sponsor, endorse, sell or promote any ETF, and makes no representation or warranty, express or implied, regarding the advisability of investing in any such ETF. In connection with the services described herein, the Company does not act as a fiduciary to any investor, fund, sponsor, issuer or other party. The Company is not registered with the U.S. Securities and Exchange Commission or any state securities authority as an investment adviser or exempt reporting adviser. The Company's role is limited to providing research and related services, and it does not provide individualized investment advice, portfolio management, brokerage services or other services requiring registration as an investment adviser.

// RELATED RESEARCH

Keep reading.

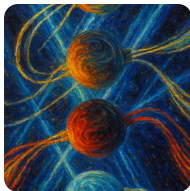
August 10, 2026

CORE RESEARCH

TPU Long and Prosper: Don't Cry Because Gemini's Over, Smile Because TPU Happened

Core Research Exclusive

By Konrad Wang and Tony Chien



August 7, 2026

AI NETWORKING MODEL

Epic Rap Battle of Connectivity: VCSELs vs MicroLEDs

By Julien Martin-Prin and Daniel Nishball

August 7, 2026

ACCELERATOR & HBM

CORE RESEARCH

MEMORY MODEL

Prefer Samsung over SK Hynix on HBM Pricing

Multi-Vertical Note

By Timothy Chang, Nick Doyle, and Dylan Patel